



Society of Petroleum Engineers

SPE-227269-MS

Benchmarking Error-Bounded Compressors for Big Seismic Data

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227269-MS](https://doi.org/10.2118/227269-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

The rapid advancement in seismic data acquisition technology, with its high-channel counts and point-source, point-receiver systems, has resulted in unprecedented volumes of seismic data. Single sensor survey can easily generate hundreds of terabytes, challenging conventional storage and data management infrastructures. Efficient compression of seismic data emerges as a critical solution to mitigate storage demands and accelerate data processing workflows. This study evaluates several state-of-the-art lossy compressors -Bitcomp, SZ3, and ZFP - focusing on their compression effectiveness and the impact of compression errors on seismic data quality. We analyze performance metrics such as compression ratio and throughput for both compression and decompression, while also examining data fidelity through PSNR (Peak Signal-to-Noise Ratio) measurements in pre-stack and post-stack domains. Our work further includes a comparison with commonly used lossless compressors, highlighting their limited compression capability for seismic datasets.

Introduction

The latest seismic acquisition practices are moving towards recording high-channel count data using smaller field arrays or single sensors, leading to better spatial sampling of seismic wavefields. These improvements yield extremely dense datasets, where trace density can exceed 100 million traces per square kilometer. As a result, the total data volume for a survey can easily reach several hundreds of terabytes, as illustrated in [Figure 1](#). The challenges extend further due to processing modules that substantially increase data volume. For example, multidimensional seismic data interpolation and regularization procedures (shown in [Figure 2](#)) can enhance final image resolution but simultaneously inflate data size dramatically. Interpolating a dataset from receiver and shot spacings of 25×150 m and 100×25 m respectively to a finer 25×25 m sampling can increase data volume from 80 terabytes per 1000 km² to over 2 petabytes per 1000 km². Such growth in

data volume severely strains disk capacity and I/O bandwidth, restricting these processes to small areas of interest rather than full surveys.

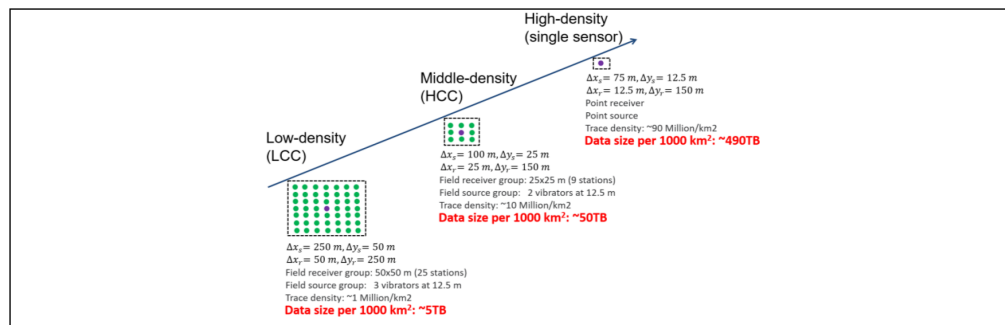


Figure 1—Evolution of seismic data acquisition: transition from low to high-density seismic survey, resulting in a huge increase in the amount of recorded seismic data per 1000 km².

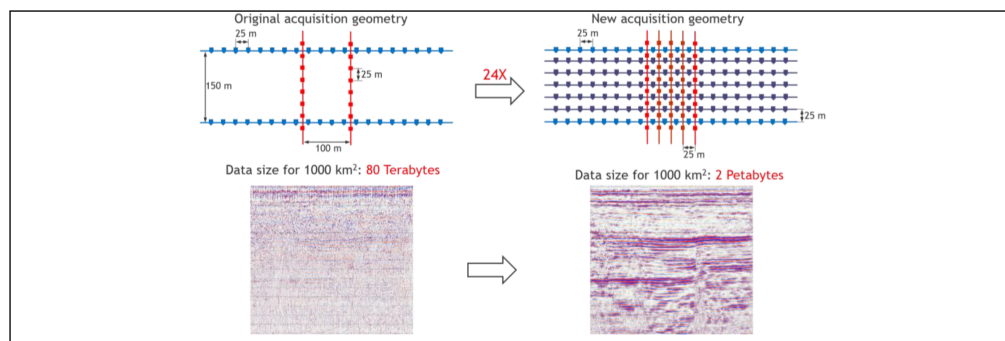


Figure 2—Example of seismic data growth during processing: interpolation from low-density acquisition to high-density geometry, resulting in a substantial increase in data size from 80 TB to 2 PB per 1000 km².

The growing data demands necessitate efficient compression techniques to reduce storage needs and improve data handling throughput. Compression algorithms can be broadly categorized into two types: lossless and lossy. Lossless compression algorithms aim to reduce the size of the data without losing any information, allowing the original data to be fully reconstructed. Although lossless compression techniques can preserve all the original data, they are not very effective for seismic datasets, typically achieving only a 5-10% reduction in size. On the other hand, lossy compression techniques provide much higher compression ratios but at the cost of some loss of information during compression. Lossy compression algorithms are designed to reduce the size of the data by discarding some of the information, while still maintaining the overall quality of the data. The main idea behind lossy compressors is to identify and remove the redundant or less important information, thereby reducing the size of the data. Lossy compressors can be categorized into two main types: error-bounded and error-unbounded compressors. Error-bounded compressors allow for control over the level of error introduced during the compression process, enabling a balance between compression ratio and error tolerance to meet specific quality standards. In contrast, error-unbounded compressors, such as fixed-rate compressors, compress data to a fixed rate regardless of its complexity, without regard for the level of error introduced. This distinction is important, as different applications and data types may require different approaches to compression, and understanding the characteristics of each type of compressor is crucial for selecting the most suitable one. Lossy compressors also vary by approach, including transform-based methods, predictive models, and quantization techniques, each with particular trade-offs.

Furthermore, lossy compression methods can be classified into local and global approaches. Local methods analyze data within small blocks or neighborhoods to optimize compression, whereas global

methods capture the overall structure of the data to achieve compression. Global compression approaches, such as TTHRESH, based on Tucker tensor decomposition, have demonstrated impressive compression ratios for some applications by capturing the global data structure (Ballester et al, 2015; Ballester et al, 2019). Although these methods are computationally heavy and may not be suitable for real-time or interactive seismic processing workflows, they could be viable options for long-term storage or backup purposes, where computational efficiency is less critical. This paper focuses primarily on local compression methods due to their efficiency and adaptability to seismic datasets, making them more suitable for large-scale seismic workflows where speed and flexibility are essential.

Our study begins by evaluating the main lossless and light lossy compressors available in the NVIDIA NVCOMP package (NVIDIA, 2022), such as Cascaded, Snappy, LZ4, Bitcomp (lossless), Gdeflate, and ZSTD. These compressors demonstrate limited performance on seismic data, with compression ratios insufficient for practical large-scale deployment. Building on this baseline, we investigate advanced lossy compressors specifically adapted to scientific data: Bitcomp, SZ3 (version 3.2.1), and ZFP (version 1.0.0). These compressors all employ local correlation detection, analyzing data within small blocks or neighborhoods to optimize compression while controlling error bounds. Bitcomp is a GPU-oriented 1D compressor developed by NVIDIA, notable for its exceptional speed with throughput exceeding 300 GB/s on GPUs during reverse time migration workflows. SZ3 applies predictive modeling with a combination of Lorenzo and regression-based predictors on blocks of data (Zhao et al., 2021; Liang et al., 2023), offering error-bounded compression with user-friendly configuration. ZFP, another popular scientific compressor, uses transform and quantization techniques to deliver controlled lossy compression with high throughput and accuracy (Lindstorm et al., 2006; Diffenderfer et al., 2019). Previous research (Dmitriev et al., 2023) has shown the effectiveness of Bitcomp, SZ3, and ZFP compressors in compressing the forward modeling seismic wavefield in Reverse Time Migration, achieving good compression ratios without compromising data quality.

We assess these compressors through several performance metrics, including compression ratio, distortion measured by PSNR in both pre-stack seismic gathers and the final stacked images, and compression/decompression speed. We also consider the performance of fixed-rate ZFP compression in relation to its error-bounded counterpart, providing a comprehensive evaluation of reconstruction quality versus compression efficiency.

Absolute error bound criteria and evaluations metrics

We employ the absolute error bound criteria (Liang et al., 2023) to control the level of error introduced by lossy compression. This criterion involves setting a limit on the maximum allowed absolute error between the original and reconstructed data, expressed as:

$$|u_i - \tilde{u}_i| \leq \delta,$$

Where u_i represents the original data sample and $\tilde{u}_i$ represents the reconstructed data sample.

Parameter δ is given by $\delta = \varepsilon r$, where ε is the user-defined relative error bound in the range of $[0, 1]$, and r is the dynamic data value range defined as $r = \max(u_i) - \min(\tilde{u}_i)$. A more precise reconstruction of the compressed data can be achieved with smaller values of ε . For lossless compressors $u_i = \tilde{u}_i$ and $\delta = 0$.

We use following performance metrics to evaluate different compressors:

Compression ratio (CR)

A compression ratio is a standard metric used to assess the efficiency of a compression algorithm. It is defined as the ratio of the size of the uncompressed data to the size of the compressed data, providing an indication of the degree of compression achieved by the algorithm.

Data distortion in pre-stack domain

To quantify the quality of data reconstruction, we utilize the Peak Signal to Noise Ratio (PSNR) metric, which is a widely accepted and standard metric for evaluating data distortion in many domains, including image and signal processing. PSNR provides a measure of the ratio between the maximum possible power of a signal and the power of corrupting noise, allowing us to assess the fidelity of the reconstructed data. A higher PSNR value indicates a better reconstruction quality, with less distortion and noise. PSNR between original trace u and reconstructed trace $\tilde{u}$ is calculated as:

$$PSNR = 20 \log_{10} \left(\frac{r}{RMSE(u, \tilde{u})} \right),$$

where r is d $RMSE$ is root mean square operator, which is defined as follows:

$$RMSE(u, \tilde{u}) = \sqrt{\frac{1}{N_t} \sum_{i=1}^{N_t} (u_i - \tilde{u}_i)^2},$$

where N_t is the number of time samples in each trace. After computing the PSNR for each trace, we calculate the mean PSNR for all traces to obtain a comprehensive measure of the reconstruction quality. Additionally, we visualize the error distribution using histograms, which facilitate the analysis of the error spread and enable us to assess the consistency and robustness of the reconstruction algorithm across different traces.

Data distortion in post-stack domain

We assess the quality of the final stacked subsurface image using the Peak Signal-to-Noise Ratio (PSNR) metric. To evaluate the impact of compression, we calculate the PSNR for stacked volumes generated from two distinct processing workflows: one without compression and another with compression. This comparison enables us to quantify the effect of compression on the final stacked image quality.

Compression and decompression speeds

Compression (decompression) speed is calculated as the ratio of the original trace size to the time taken for compression (decompression). We record the compression and decompression time for each trace and then determine the overall compression and decompression throughput for the complete dataset. These throughput metrics are critical for practical deployment, especially when handling terabyte-scale datasets requiring efficient I/O operations.

Experimental setup and results on field data

To demonstrate the compression performance of our approach, we utilize 3D high-channel count land seismic data. Specifically, we select a single data line consisting of around seven million of seismic traces, approximately 35 GB in size, arranged in CMP-offset order. We apply a standard CMP processing sequence to prepare the data for stacking and then apply data compression to the processed data. In our tests, we focus on 1D compression, where compression is performed on a trace-by-trace basis, solely in the time dimension. Although multidimensional compression can yield higher compression ratios (e.g., by considering both time and offset dimensions), we adopt a trace-by-trace approach due to the significant variability in amplitude ranges among seismic traces within a gather, particularly in raw data. This variability, often caused by background noise resulting in extremely high-amplitude noisy traces, makes it challenging to set a maximum allowed compression ratio for the ensemble of traces.

For lossless compression, the results are shown in [Table 1](#), where one can see that the maximum compression ratio is achieved using the ZSTD compressor, with a compression ratio of 1.45038 for processed data. This high compression ratio is likely due to the fact that the processing workflow included a mute step, which introduced many zero values in each trace, making it easier to compress. In contrast, the results for raw data are shown in [Table 2](#), where the maximum compression ratio is only 1.07206, also

achieved by the ZSTD compressor. Overall, the results suggest that lossless compressors are not effective in achieving significant compression ratios, particularly for raw data.

Table 1—Lossless compression ratios for processed seismic data.

Compressor	Compression ratio
Cascaded	1.09181
Snappy	1.31869
LZ4	1.33574
Bitcomp	1.33602
Gdeflate	1.4332
ZSTD	1.45038

Table 2—Lossless compression ratios for raw seismic data.

Compressor	Compression ratio
Cascaded	0.994174
Snappy	0.991963
LZ4	0.99585
Bitcomp	0.999021
Gdeflate	1.06793
ZSTD	1.07206

In [Table 3](#), the total compression ratios for all error-bounded compressors are presented for different relative error bound ratios ε . It can be observed that Bitcomp consistently achieves the highest compression ratios across all values of epsilon, making it a strong performer in terms of compression efficiency. ZFP also exhibits competitive compression ratios, although slightly lower than those of Bitcomp. Meanwhile, in 1D mode, the compression ratio of SZ3 for small values of ε ($\varepsilon > 1.0e^{-3}$) is significantly lower, indicating that it may not be the optimal choice for this specific scenario. Later in the paper, we will show that the SZ3 compressor was operated in lossless mode for $\varepsilon = 1.0e^{-4}$ and $\varepsilon = 1.0e^{-5}$. It is worth noting that our evaluation of SZ3 was limited to the case where the compression algorithm was set to ALGO_LORENZO_REG, which utilizes Lorenzo and regression-based predictors for data compression. Other algorithms within the SZ3 framework may potentially offer better compression performance. [Table 4](#) presents the total compression ratio for the ZFP compressor in fixed rate mode. The results show that the actual compression ratio is close to the desired ratio, with values of 7.911, 5.296, 3.980, and 1.996 for rates of 4, 6, 8, and 16, respectively. This indicates that the compressor effectively achieves the desired compression level, with a high degree of accuracy.

Table 3—Overall compression ratio for different error-bounded lossy compressors for processed seismic data.

Compressor	Relative error $\varepsilon=1.0e^{-1}$	Relative error $\varepsilon=1.0e^{-2}$	Relative error $\varepsilon=1.0e^{-3}$	Relative error $\varepsilon=1.0e^{-4}$	Relative error $\varepsilon=1.0e^{-5}$
Bitcomp	12.2898	6.425	4.265	3.183	2.539
SZ3	10.2308	4.4289	1.66189	1.369	1.367
ZFP	6.83441	4.487	3.338	2.657	2.206

Table 4—ZFP compression ratios with fixed-rate mode for processed seismic data.

Compressor	Rate 4	Rate 6	Rate 8	Rate 16
ZFP	7.911	5.296	3.980	1.996

Setting the same epsilon value does not necessarily imply the same reconstruction data quality. To investigate this further, we examined the distribution of PSNR values for different relative error bounds. Figure 4 illustrates the error distribution between original and reconstructed seismic traces, computed across the entire pre-stack dataset (approximately 7 million seismic traces). The results show that the mean PSNR values vary substantially among different compressors and relative error bounds. Notably, ZFP consistently exhibits the highest mean PSNR among all compressors for each value of ϵ , indicating its superior reconstruction quality. Both Bitcomp and ZFP provide excellent quality at $\epsilon = 1.0e^{-5}$ (with mean PSNR values of 112.126 for Bitcomp and 119.79 for ZFP). Furthermore, we observe that SZ3 switches to lossless mode at $\epsilon = 1.0e^{-5}$, as evident in Figure 4f, where the mean PSNR is infinite, indicating lossless compression. This behavior suggests that the default lossy compression parameters used in SZ3 may not be optimal for this specific use case, and further tuning may be necessary to achieve effective lossy compression.

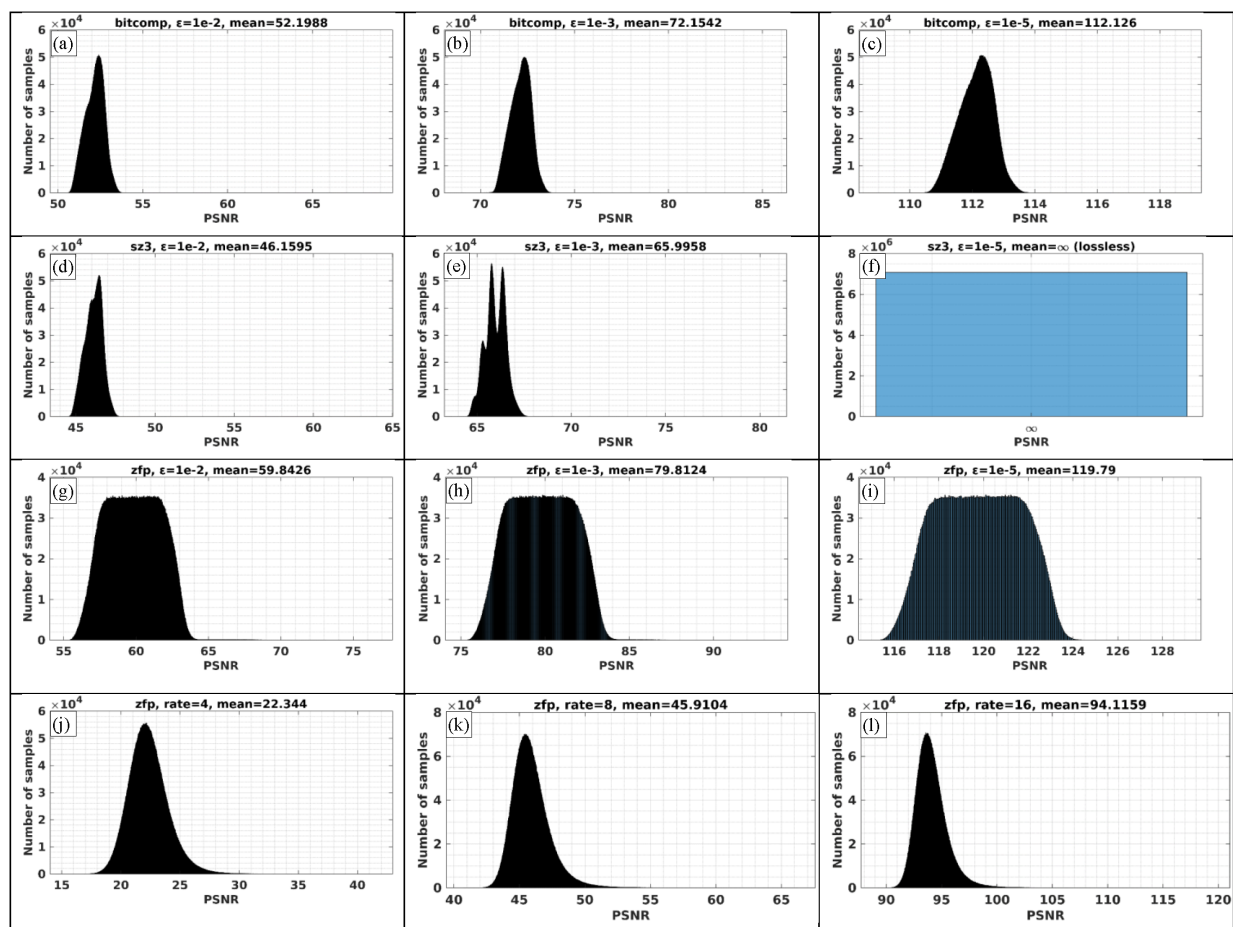


Figure 3—Comparison of PSNR error distribution between the original and reconstructed prestack data for different error-bounded and fix-rate compressors. The histogram is computed in a trace-by-trace manner using approximately 7 million seismic traces (33 GB).

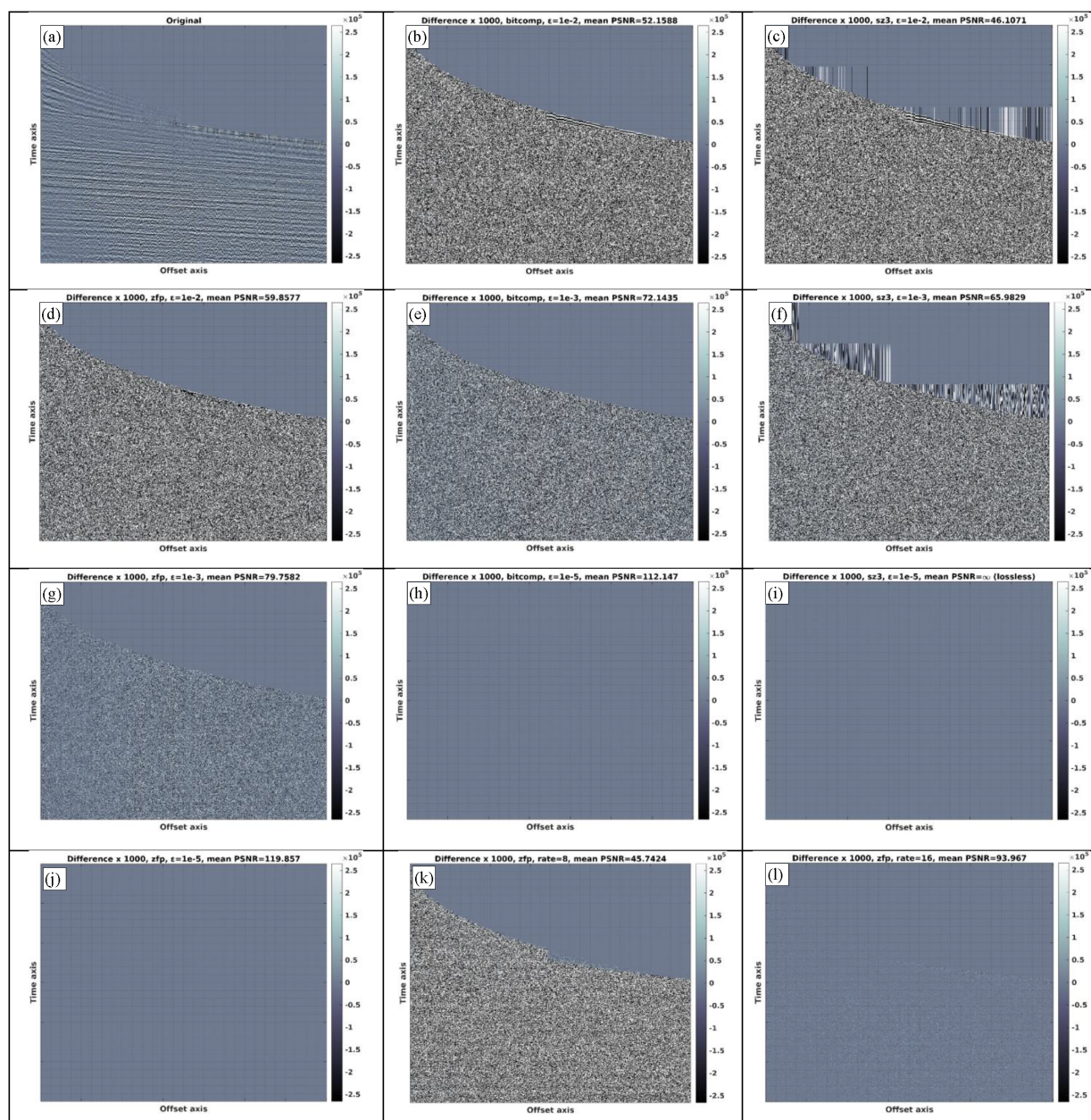


Figure 4—Effects of compression on pre-stack seismic data quality for randomly chosen CDP gather. PSNR displayed in the figures are computed within current CDP gather.

A comparative analysis of the fixed-rate zfp compressor with error-bounded compressors, considering similar compression ratios, provides further insights. For instance, at $\epsilon = 1.0e^{-2}$, bitcomp achieves an overall compression ratio of 4.265 and a mean PSNR of 52.1988, whereas zfp's fixed-rate compression at a rate of 8 yields a CR of 3.980 and a mean PSNR of 45.9104. This shows that error-bounded compressors can offer better compression ratios and higher mean PSNR values than fixed-rate compressors. The quality of reconstruction improves with higher rates for fixed-rate compressors, as observed in Figure 31, where zfp at a rate of 16 achieves a CR of 1.996 and a mean PSNR of 94.1159. However, error-bounded compressors continue to outperform fixed-rate compressors in terms of both compression ratio and mean PSNR. For example, at $\epsilon = 1.0e^{-5}$, Bitcomp achieves a CR of 2.539 and a mean PSNR of 112.126, surpassing the performance of zfp's fixed-rate compression.

Next, we visually assess the error introduced by compression in pre-stack data. Ideally, the error should be of a random nature, as we perform extensive stacking operations in seismic processing workflow, and

coherent error stacking is undesirable. Figure 4 illustrates a randomly selected original CDP gather and the error between the original and reconstructed gather for different compressors and different error bounds. To facilitate comparison, all figures are displayed on the same scale, while the error difference is amplified by a factor of 1000 to reveal its behavior. Our analysis reveals that the error is predominantly random in all cases. However, some coherent data leakage occurs for larger epsilon values (Figure 5b, 5c, and 5d), particularly in the boundary region separating values close to zero and non-zero values, which corresponds to the mute applied during processing. Notably, the SZ3 compressor introduces vertical stripes (Figure 5c and 5f) by replacing zero values above this boundary with small non-zero values, which could lead to coherent error stacking if not these artifacts are not muted before stacking. The results on this figure confirm our previous findings that ZFP gives the highest PSNR values among all relative error bounds.

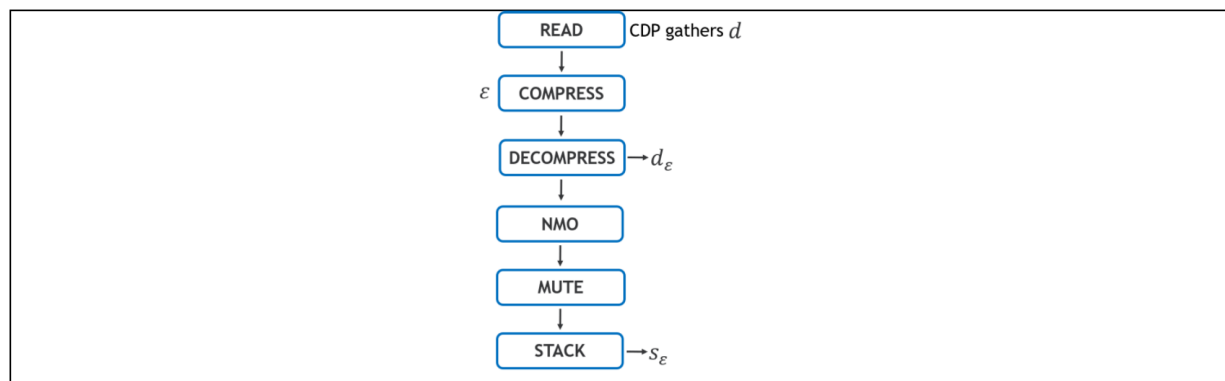


Figure 5—Workflow for evaluation the effect of compression induced error on the final seismic stacked image.

The ultimate objective of our processing workflow is to produce a high-quality stacked seismic image. As illustrated in Figure 5, the standard processing workflow involves reading processed data, applying Normal Moveout Corrections (NMO), mute, and stacking traces within Common Depth Point (CDP) gathers. To investigate the impact of compression errors on the stacked image, we examine the sensitivity of these errors to the stacking process. Specifically, we consider scenarios where multiple seismic traces with small, coherent errors introduced by lossy compressors are stacked, potentially resulting in a noisy stack due to the cumulative effect of these errors. Even small coherent events can amplify into significant errors after stacking, highlighting the importance of minimizing compression errors. Ideally, a random error distribution is desirable, as it has a negligible impact on subsequent geological interpretations. To evaluate the effects of compression, we perform the stacking workflow (Figure 5) using both original data (without compression) and reconstructed data at various relative error bounds (ϵ). We then compare the Peak Signal-to-Noise Ratio (PSNR) values between the original stack (s_0) and the stacks obtained with compression (s_ϵ), allowing us to assess the impact of compression on image quality.

Figure 6 displays the original stacked image, as well as the difference between the original stacked image and the stacked images obtained using various lossy compressors with different relative error bounds (ϵ) for error-bounded compressors and different compression rates for the fixed-rate zfp compressor. To highlight the error distribution, we scale the difference by a factor of 1000, following the same procedure as before. We observe a consistent trend in terms of PSNR for post-stack data and pre-stack gathers for both Bitcomp and ZFP. Specifically, at a relative error bound of $\epsilon = 1.0e^{-5}$, Bitcomp achieves a mean PSNR of 112.126 for pre-stack data (Figure 3c) and a similar mean PSNR of 113.981 for the stacked image (Figure 6h). Similarly, ZFP exhibits a mean PSNR of 119.79 for pre-stack gathers (Figure 3i) and a mean PSNR of 120.77 for the stacked image (Figure 6j). For both Bitcomp and ZFP, a small part of the stack shows coherent data leakage, specifically a rectangular area located in the middle in the horizontal direction and the deep part in time. This leakage nearly disappears at $\epsilon = 1.0e^{-5}$ (Figures 6h and 6j). SZ3 appears to be more sensitive to the

stacking operation, with a notable decrease in PSNR for the stacked image compared to pre-stack gathers. For instance, at $\varepsilon = 1.0e^{-3}$, the mean PSNR for pre-stack gathers is 65.9958, whereas the mean PSNR for the stacked image is 49.3957. This sensitivity may be attributed to the "vertical" stripes behavior exhibited by SZ3, as illustrated in Figure 5c. Additionally, as previously noted, SZ3 switches to lossless mode for small values of ε (e.g., $1.0e^{-4}$ and $1.0e^{-5}$), resulting in infinite PSNR values. For the fixed-rate ZFP compressor, a coherent error leakage is visible at a rate of 8 among the stack section (Figure 6k, mean PSNR = 56), whereas a rate of 16 yields a more accurate stack (Figure 6l, mean PSNR = 104.265).

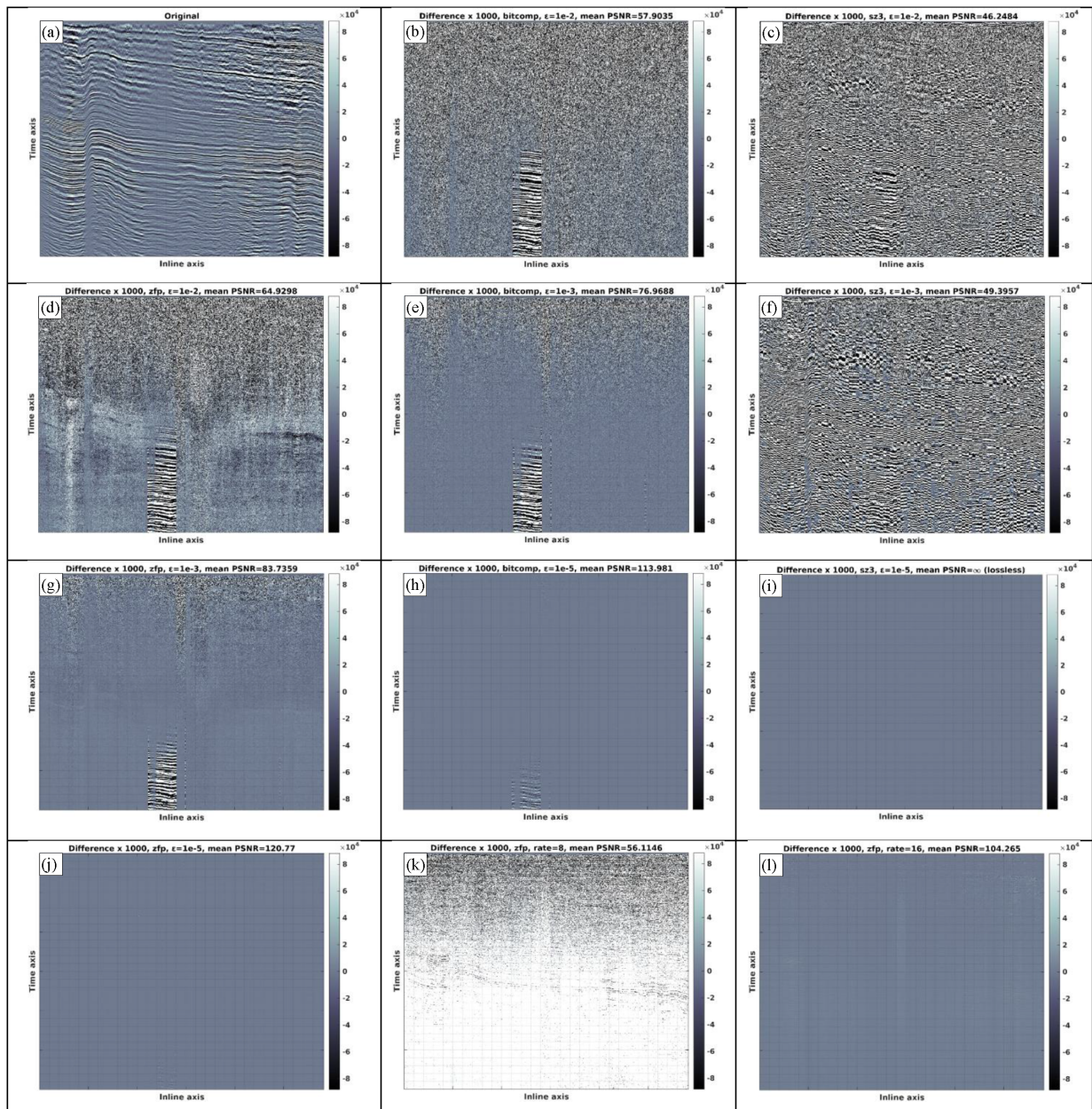


Figure 6—Effects of compression on seismic stack quality: a comparison of PSNR error distributions using original and compressed data at different relative error bounds and bit rates, processed data.

Figure 7 presents a histogram of error distributions between the original stack and the stack obtained using decompressed data at different relative error bounds and rates. The results reveal that the range of PSNR values around the mean is relatively large for error bounded compressors compared to the pre-stack error distribution, indicating that some traces have poorer reconstruction quality. For instance, Bitcomp has

a mean PSNR of 113.981 at $\epsilon = 1.0\text{e-}5$, but a small number of traces in the stack are reconstructed with a PSNR as low as 60 (Figure 7c). Similar behavior is observed for the ZFP error-bounded compressor (Figure 7i). In contrast, ZFP at a rate of 16 exhibits consistent results with a narrow error distribution around the mean, similar to the pre-stack data but with much lower CR.

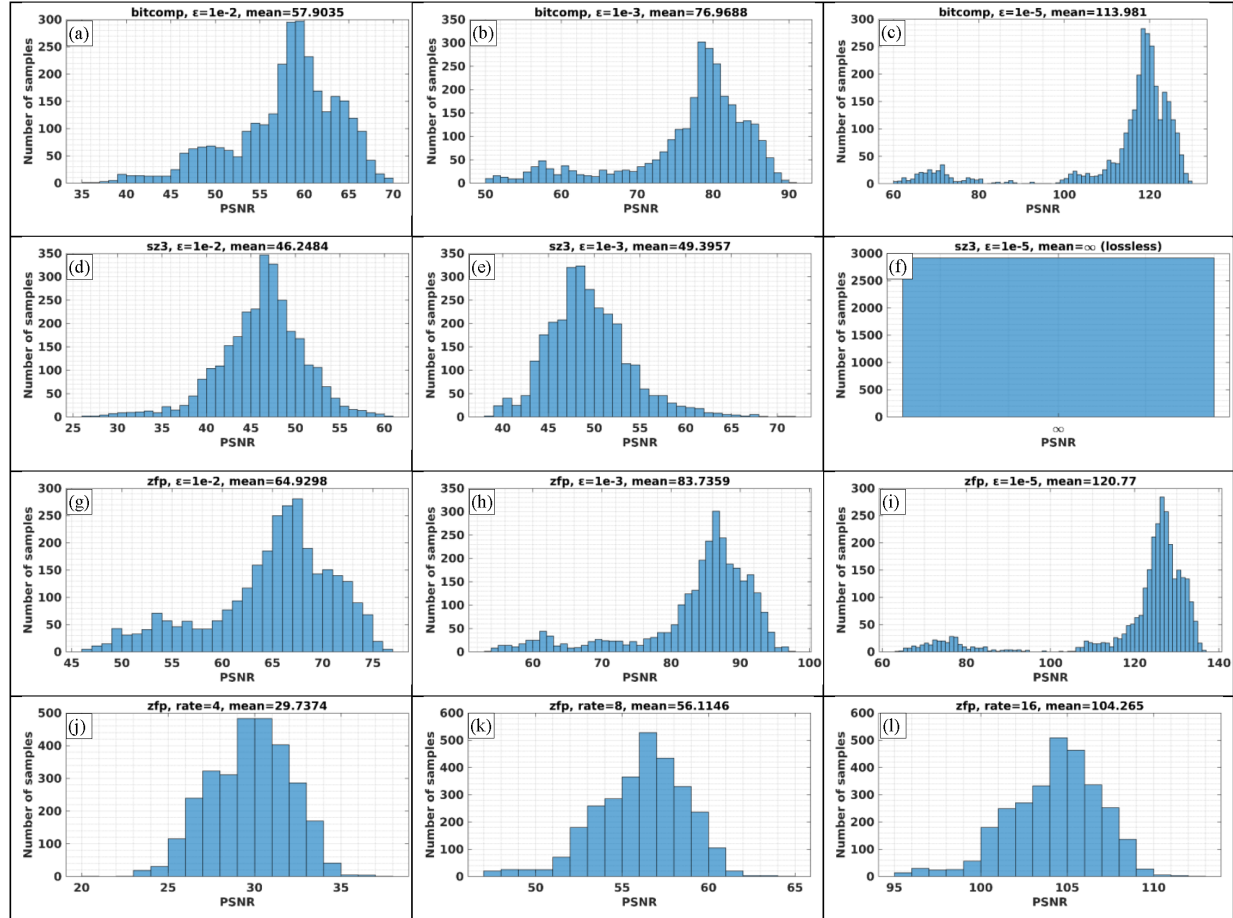


Figure 7—Comparison of PSNR error distributions between original stacked image and image obtained with lossy compressors at different relative error bounds and bit rates.

Finally, we assess the compression and decompression throughput of various compressors, as this is a crucial factor when working with large datasets. All computations are performed in 32-bit single precision on a single node of a supercomputing cluster, which is equipped with a dual-socket 20-core Intel Xeon Gold 6248 CPU and 384 GB of DRAM. The compression and decompression throughputs for all compressors are presented in Table 5, using one-line seismic data as a benchmark (7 million seismic traces). Note that the zfp compressor is executed in parallel using OpenMP, whereas the SZ3 compressor is run without OpenMP, which may potentially limit its throughput. In contrast, the Bitcomp compressor, which is primarily optimized for GPU acceleration and does not support OpenMP configuration, is run in its native single-threaded mode on the CPU. Despite this, Bitcomp achieves the highest compression and decompression throughput among all compressors, with compression throughput ranging from 343 to 363 MB/s and decompression throughput ranging from 1393 to 1481 MB/s.

Table 5—Overall compression / decompression speed (MB/s) at different relative error bounds.

Compressor	Relative error $\varepsilon=1.0e^{-2}$	Relative error $\varepsilon=1.0e^{-3}$	Relative error $\varepsilon=1.0e^{-4}$	Relative error $\varepsilon=1.0e^{-5}$
Bitcomp	363.8 / 1481.2	351.5 / 1431.1	346.4 / 1409.3	343.9 / 1393.2
SZ3	59.2 / 83.4	26.5 / 64.8	11.1 / 494.4	5.1 / 428.5
ZFP	205.6 / 150.8	197.9 / 117.1	189.6 / 98.5	182.6 / 82.4

Conclusion and discussions

This study presents a comprehensive analysis of leading lossless, error-bounded, and fix rate compressors for seismic data compression, evaluating their performance based on various metrics. The impact of compression errors on data reconstruction quality was assessed in both pre-stack and post-stack domains, with data quality measured using PSNR values. Although lossless compression is often preferred for its ability to preserve data integrity, our results indicate that it may not be the most effective approach for seismic data compression. This is evident from the fact that the best lossless compressor achieved a maximum compression ratio of only 1.07206 for raw data, highlighting the limitations of lossless compression for seismic data. In contrast, we demonstrate that moderate reductions in data size can be achieved without sacrificing data quality in both pre-stack and post-stack domains. For instance, at a relative error bound of $\varepsilon = 1.0e^{-5}$, both Bitcomp and ZFP exhibit good data quality and moderate compression rates. Specifically, Bitcomp achieves a compression ratio of 2.5, with mean PSNR values of 112.126 and 113.981 for pre-stack and post-stack data, respectively. Similarly, ZFP achieves a compression ratio of 2.206, with mean PSNR values of 119.79 and 120.77 for pre-stack and post-stack data, respectively. Furthermore, the fixed-rate ZFP compressor yields good data reconstruction quality at a rate of 16, with a compression ratio of 1.996 and mean PSNR values of 94.1159 and 104.265 for pre-stack and post-stack data, respectively.

Notably, Bitcomp exhibits the highest compression and decompression throughput among all compressors, with compression speeds ranging from 343 to 363 MB/s and decompression speeds ranging from 1393 to 1481 MB/s, making it a suitable choice for applications requiring fast data processing. Furthermore, the performance of Bitcomp can be substantially enhanced by dividing the data into smaller chunks and leveraging parallel processing using OpenMP, offering a promising avenue for further optimization.

However, determining optimal relative error bounds for all situations remains a challenging task. Seismic traces can exhibit vastly different dynamic value ranges, and using small relative error bounds (e.g., $1e^{-5}$) may be necessary to ensure data quality. While better compression ratios may be achievable for specific workflows, the impact of compression on data quality must be carefully evaluated in each case. Future research directions may include exploring multidimensional compression techniques, although defining relative error bounds within an ensemble of traces poses a significant challenge. Ultimately, our study provides valuable insights into the performance of error-bounded compressors for seismic data compression, highlighting the need for careful consideration of data quality and compression ratios in various applications.

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